

• BROWSER • RECON • HTTP TESTING • SANDBOXING

The Red-Teaming Browser for AI Agents.

- ▶ Automate red teaming. Browse applications, discover endpoints, and test HTTP traffic.
- ▶ Restrict agent access. Set network and filesystem limits for the testing workflow.
- ▶ Review the evidence. Inspect browser actions, captured traffic, and blocked connections.

Pure Rust • No Chromium or V8 • Apache 2.0 • local-first, no hosted service

■ THE PROBLEM

Automating red teaming safely is hard.

01 · Slow interfaces

Human-facing tools slow agents down

Complex GUIs and heavyweight browsers add latency and memory costs to repeated tests.

02 · Manual workflows

Red-teaming tools are hard to automate

Tools built around human operators leave agents coordinating browser state, proxies, and request editors.

03 · Scope drift

AI agents can attack the wrong systems

A mistake or prompt injection can redirect testing toward unrelated applications or internal services.

04 · Incomplete records

AI actions are hard to audit

A final report alone does not show the requests sent, responses received, or connections denied.

■ WHAT H5I GIVES YOU

Automate browsing. Control HTTP traffic. Sandbox the workflow.

`h5i browser`

Automated browsing

Navigate pages, forms, and authenticated sessions through a CLI built for agents.

`h5i recon`

Application reconnaissance

Extract and organize endpoints with observations tied to browser traffic.

`h5i websec`

Direct HTTP control

Capture, edit, replay, and compare requests in the same browser session.

`h5i box · h5i browser audit`

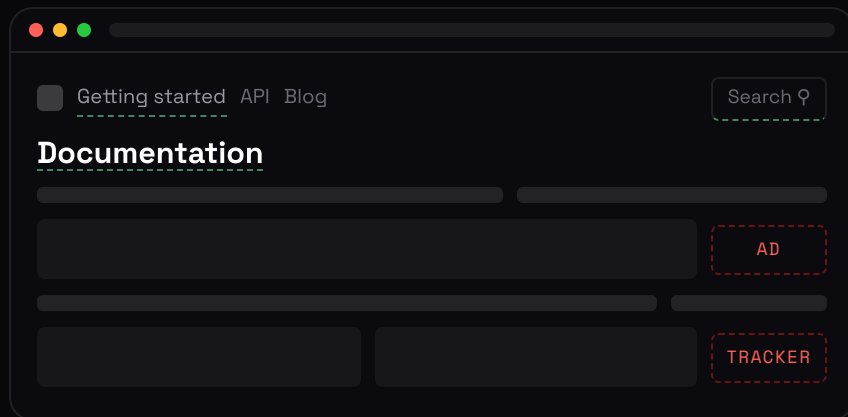
Scoped, reviewable execution

Configure isolation and access policies, then review browser actions and denied connections.

■ THE BROWSER

Give AI agents a fast browser interface.

THE WEB · WHAT A PAGE LOADS



PAGE BYTES

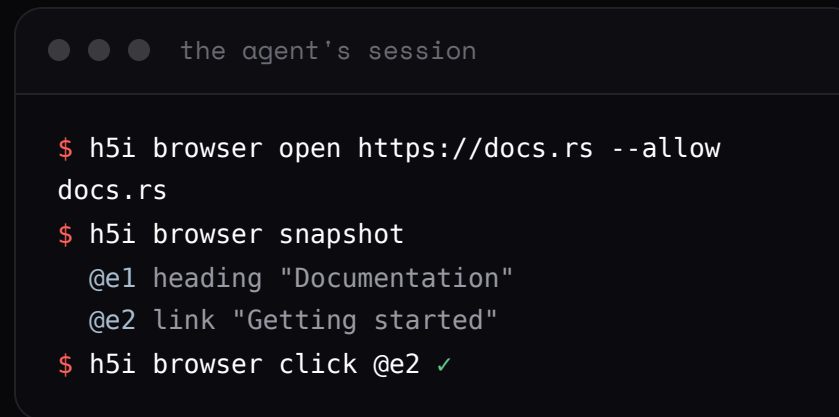
POLICY-CONTROLLED ·
SANDBOXABLE

h5i browser



OUTLINE, NOT PIXELS

THE AGENT · WHAT IT RUNS, AND WHAT COMES BACK



* Simple-site benchmarks vs headless Chromium · see the browser design notes

~3x

faster reads*

~86%

less peak memory*

Pure Rust

no Chromium or V8

CLI

structured page output

■ RECON PLUGIN

Discover application endpoints. Keep their evidence.

The optional recon plugin uses browser sessions and network policy for application reconnaissance.

```
h5i recon extract · known · import
```

Collect endpoint candidates

Extract URLs from HTML, JavaScript, JSON, and headers. Add known paths or import an existing inventory.

```
h5i recon endpoints
```

Track what was observed

Separate discovered candidates from observed, confirmed, and refused endpoints. Keep identities distinct.

```
h5i recon triage
```

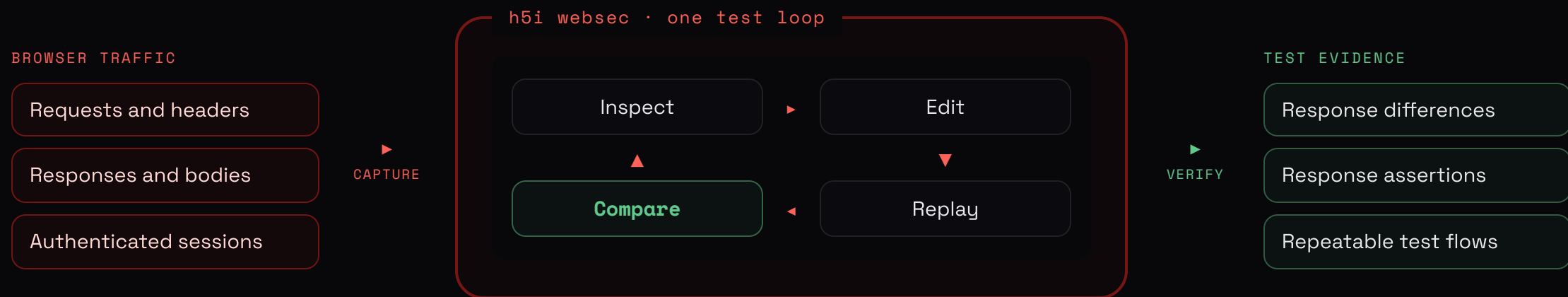
Review the supporting traffic

Trace observations to browser message IDs. A confirmed endpoint is not a confirmed vulnerability.

■ WEB SECURITY TESTING

Inspect and test web apps from **the same browser**.

The optional websec plugin lets agents capture, edit, replay, and compare browser traffic. No separate MITM proxy, CA setup, or repeater.



Browse the app. **Test its traffic.** Re-run the flow for **authorized security testing**.

Set the agent's access limits.

Review what it did.

Browser policy covers browser traffic. To constrain other tools, sandbox the whole workflow; enforcement depends on the isolation tier.

`.h5i/env.toml`

Restrict destinations

Allow the application and services in scope. Browser policies reject other destinations and record denied attempts.

`h5i box`

Isolate the testing workflow

Run the agent and its tools inside a sandbox with explicit network and filesystem limits.

`h5i browser audit · requests`

Inspect the execution record

Review browser actions, HTTP traffic, and denied connections with their reasons.

`h5i ui`

Review and intervene

Inspect sessions and take over sensitive browser steps when human input is needed.

■ REPOSITORY MILESTONES

An open-source foundation to build on.

600+

GitHub stars

60+

forks

20+

contributors

20+

published releases

GitHub · checked September 10, 2026 · [Repository and releases](#)

■ INITIAL USERS

For teams building AI red-teaming workflows.

Security engineers

Automate authorized testing

Bring application browsing, reconnaissance, and HTTP testing into agent-driven security work.

Agent developers

Build on one browser interface

Use consistent session state and CLI commands across browser and plugin capabilities.

Security researchers

Make experiments reviewable

Use controlled environments for CTFs and research, with traffic evidence and execution records.

■ WHY h5i

Built for AI red teaming.

Fewer tools to coordinate.

Conventional browser automation

Broad website compatibility and mature automation. HTTP security testing adds a separate workflow.

Browser + proxy tools

Rich traffic inspection, with browser state, proxy setup, and test scripts to coordinate.

Sandboxing an existing browser

Constrains execution and access, but does not itself add HTTP inspection or replay.

h5i

A pure-Rust browser combining page automation, recon, HTTP testing, network policies, and configurable isolation.

Website compatibility remains a tradeoff: h5i is not a full Chromium replacement.

■ NEXT MILESTONES

Make h5i the browser AI red teams build on.

Product

Broaden coverage

Improve website compatibility and repeatable browser, recon, and HTTP testing workflows.

Adoption

Work with design partners

Target 10 design partners building AI red-teaming workflows; work toward the first paid pilots.

Open source

Build with the community

Keep improving a local-first, Apache 2.0 browser alongside contributors and security researchers.

h5i.dev

github.com/h5i-dev/h5i